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Predicting Hospitalization & Mortality Using the Electronic Frailty Index

Biostatistics Internship Project

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Background: The Frailty Problem

Why Frailty Matters

- Aging population: $\sim 16\%$ of adults ≥ 65 are frail
- Potential to predict adverse outcomes independent of age and comorbidity
- Electronic Frailty Index (eFI) derived from EHR data

Study Goal

Replicate & extend Pajewski et al. (2019) using a **diagnosis-based eFI** from EHR/claims data.

Research Question

Can an eFI computed from routinely collected EHR variables predict hospitalization and mortality?

Research Objectives

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1. **Describe baseline characteristics** by frailty category
2. **Test predictive ability** of eFI and 36 clinical components
3. **Compare performance** to prior literature

Outcomes Studied

Outcome	Period
Hospitalization	6 months, 12 months
Mortality	6 months, 12 months

Electronic Frailty Index: Concept & Calculation

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Deficit Accumulation Theory
Frailty reflects cumulative health deficits across multiple systems. Each deficit adds to overall vulnerability.

- Predicts hospitalization, mortality, care needs
- Derived from pre-existing records

eFI Calculation

- Diagnosis-based (ICD codes, labs, meds)
- 36 clinical deficits counted
- eFI = (deficits present) / (36 total)
- Range: 0 (fit) to 1 (most frail)

eFI Categories

Metric	Fit	Pre-Frail	Frail
eFI cutoff	≤ 0.10	0.10–0.21	> 0.21
Our sample	19.4%	47.9%	32.6%

Baseline Characteristics by Frailty Category

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Baseline Characteristics by Frailty Category

Mean ± SD or N (%)

Characteristic	Fit	Pre-Frail	Frail	P-value
N	3,554	8,768	5,956	
Age (years)	66.4 ± 6.8	71.2 ± 6.5	76.2 ± 6.1	<0.001
Female (%)	48.2	52.1	57.3	<0.001
CCI Score	2.1 ± 1.8	3.2 ± 2.1	4.8 ± 2.4	<0.001
eFI Score	0.07 ± 0.02	0.15 ± 0.03	0.28 ± 0.06	<0.001

What This Shows

- **Age**: Mean increases from ~66 (Fit) to ~76 (Frail)
- **CCI**: Comorbidity burden rises from ~2 (Fit) to ~5 (Frail)
- **eFI score**: Deficits accumulate as expected across categories

All inter-group comparisons: $p < 0.001$ (Kruskal-Wallis / χ^2)

Key: This validates that Fit/Pre-Frail/Frail represent distinct clinical populations.

Methods: Advanced Statistical Approach

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Firth's Penalized Logistic Regression

Why we used it:

- Only 6 deaths at 6 months, 14 at 12 months
- Standard logistic regression fails with rare events (separation issues)
- Firth's method applies penalized likelihood correction for stable estimates
- Implemented via `logistf` package

Output: ORs, 95% CIs, and p-values directly from penalized fit

Restricted Cubic Splines (RCS)

Why we used it:

- Allowed non-linear relationship between continuous eFI and outcomes
- 5 knots placed at standard percentiles
- Implemented via `rms` package with `datadist`

Output: Predicted probabilities at key percentiles (10th, 25th, 50th, 75th, 90th, 95th)

Study Population

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Cohort Flow

Initial $N = 24{,}929$

↓

Missing eFI: 6,651 (26.7%)

↓

Analytic sample = 18,278

Selection Bias Check

Excluded patients were **significantly older**:

Selection Bias Check

Group	Age	p
Included	70.8 ± 7.2	
Excluded	72.7 ± 6.9	< 0.001

Complete-case analysis adopted (no imputation).

Results: Modest Discrimination for Hospitalization

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Overall eFI Performance

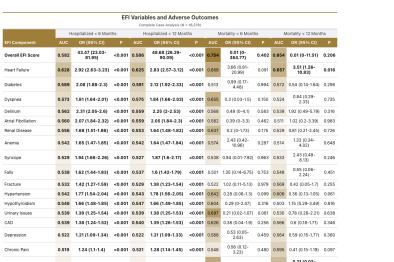
Hospitalization

- 6-month AUC = **0.582**
- 12-month AUC = **0.586**

Interpretation: Barely better than random chance (AUC = 0.50).

Contrast with Literature

Pajewski et al. reported higher AUC (≈ 0.65).
Likely reason: They included utilization variables (prior ED visits, admissions) that are stronger predictors of future hospitalization.



eFI Variables and Adverse Outcomes

Complete Case Analysis (N = 16,219)

eFI Component	Hospitalized < 6 Months			Hospitalized < 12 Months			Mortality < 6 Months			Mortality < 12 Months		
	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P
Overall eFI Score	0.582	**43.67 (22.03–87.09)**	**<0.001**	0.586	**48.66 (28.29–85.09)**	**<0.001**	0.760	**0.01 (0–0.01)**	0.402	**0.854**	**0.01 (0–11.01)**	0.208
Heart Failure	0.518	**2.92 (2.63–3.23)**	**<0.001**	0.525	**2.83 (2.57–3.12)**	**<0.001**	0.650	7.86 (0.01–20.96)	0.081	**0.657**	**3.51 (1.26–10.83)**	**0.010**
Diabetes	0.509	**2.08 (1.88–2.3)**	**<0.001**	0.505	**2.13 (1.92–2.33)**	**<0.001**	0.519	1.59 (0.17–4.46)	0.894	0.572	0.54 (0.14–1.84)	0.298
Dyspnea	0.573	**1.81 (1.64–2.01)**	**<0.001**	0.575	**1.84 (1.66–2.03)**	**<0.001**	0.650	5.3 (0.03–1.5)	0.156	0.524	0.84 (0.28–2.35)	0.795
Delirium	0.562	**2.31 (2.05–2.6)**	**<0.001**	0.559	**2.25 (2–2.53)**	**<0.001**	0.566	3.48 (0–4.1)	0.583	0.539	1.92 (0.49–6.79)	0.316
Atrial Fibrillation	0.560	**2.07 (1.84–2.32)**	**<0.001**	0.559	**2.05 (1.84–2.3)**	**<0.001**	0.582	3.29 (0–3.3)	0.462	0.511	1.02 (0.2–3.39)	0.983
Renal Disease	0.598	**1.68 (1.51–1.86)**	**<0.001**	0.593	**1.66 (1.48–1.82)**	**<0.001**	0.637	3.2 (0–1.73)	0.175	0.629	0.81 (0.21–2.49)	0.726
Anemia	0.542	**1.85 (1.67–1.05)**	**<0.001**	0.542	**1.84 (1.67–1.04)**	**<0.001**	0.574	2.43 (0.42–10.96)	0.287	0.514	1.03 (0.34–4.02)	0.648
Syncope	0.529	**1.56 (1.48–2.26)**	**<0.001**	0.527	**1.57 (1.8–2.17)**	**<0.001**	0.563	0.96 (0.01–7.82)	0.963	0.533	2.43 (0.48–9.13)	0.246
Falls	0.536	**1.82 (1.66–1.03)**	**<0.001**	0.537	**1.6 (1.42–1.79)**	**<0.001**	0.551	1.35 (0.14–4.75)	0.753	0.548	0.95 (0.26–2.24)	0.451
Fracture	0.532	**1.43 (1.27–1.59)**	**<0.001**	0.529	**1.39 (1.22–1.54)**	**<0.001**	0.522	1.02 (0.11–5.5)	0.979	0.508	0.42 (0.05–1.7)	0.255
Hypertension	0.562	**1.77 (1.54–2.04)**	**<0.001**	0.543	**1.78 (1.55–2.05)**	**<0.001**	0.642	0.28 (0.06–1.3)	0.099	0.606	0.36 (0.13–1.01)	0.061
Hypothyroidism	0.546	**1.66 (1.50–1.85)**	**<0.001**	0.547	**1.66 (1.49–1.85)**	**<0.001**	0.604	0.29 (0–2.47)	0.316	0.503	1.15 (0.39–3.49)	0.816
Urinary Issues	0.539	**1.39 (1.25–1.54)**	**<0.001**	0.539	**1.35 (1.21–1.53)**	**<0.001**	0.618	6.21 (1.02–1.87)	0.061	0.518	0.70 (0.28–2.21)	0.638
CAD	0.528	**1.38 (1.24–1.53)**	**<0.001**	0.540	**1.39 (1.26–1.53)**	**<0.001**	0.566	0.38 (0.04–1.9)	0.266	0.566	0.8 (0.16–1.71)	0.749
Depression	0.522	**1.21 (1.09–1.34)**	**<0.001**	0.522	**1.21 (1.09–1.33)**	**<0.001**	0.550	0.53 (0.05–2.65)	0.459	0.504	0.59 (0.15–1.77)	0.360
Chronic Pain	0.519	**1.24 (1.11–1.4)**	**<0.001**	0.521	**1.28 (1.14–1.43)**	**<0.001**	0.546	0.96 (0.12–5.23)	0.480	0.999	0.41 (0.15–1.18)	0.097
Rheumatoid Arthritis	0.526	**1.25 (1.13–1.39)**	**<0.001**	0.526	**1.25 (1.13–1.39)**	**<0.001**	0.578	0.14 (0–1.22)	0.083	**0.638**	**0.21 (0.03–0.95)**	**0.026**
Dizziness	0.519	**1.24 (1.1–1.39)**	**<0.001**	0.521	**1.26 (1.12–1.41)**	**<0.001**	0.619	0.26 (0–2.23)	0.270	0.577	0.36 (0.04–1.66)	0.262
Valvular Disease	0.519	**1.41 (1.22–1.62)**	**<0.001**	0.517	**1.38 (1.19–1.58)**	**<0.001**	0.555	0.62 (0–5.23)	0.725	0.555	0.28 (0–2.37)	0.269
Cancer (Other)	0.515	**0.99 (0.76–0.94)**	**0.008**	0.516	**0.85 (0.76–0.94)**	**0.004**	0.647	0.77 (0.09–3.96)	0.772	0.668	0.56 (0.11–1.99)	0.381
Osteoporosis	0.518	**0.72 (0.61–0.84)**	**<0.001**	0.519	**0.71 (0.6–0.82)**	**<0.001**	0.566	0.49 (0–4.13)	0.589	0.568	0.22 (0–1.64)	0.174
Weight Loss	0.516	**0.82 (0.72–0.93)**	**0.002**	0.518	**0.78 (0.68–0.88)**	**<0.001**	0.603	0.3 (0–2.58)	0.332	0.518	0.79 (0.15–2.62)	0.721
Stroke	0.513	**1.10 (1.03–1.36)**	**0.021**	0.510	**1.19 (1.03–1.36)**	**0.019**	0.567	0.6 (0–4.19)	0.596	0.567	0.22 (0–1.66)	0.179
Peptic Ulcer	0.506	**1.44 (1.1–1.90)**	**0.008**	0.506	**1.45 (1.12–1.88)**	**0.006**	0.519	2.56 (0.02–21.67)	0.724	0.521	3.7 (0.4–15.75)	0.199
Peripheral Vascular Disease	0.513	**1.13 (1.01–1.25)**	**0.028**	0.513	**1.12 (1.01–1.25)**	**0.027**	0.573	0.81 (0.06–3.04)	0.572	0.548	0.69 (0.17–2.06)	0.511
Incontinence	0.507	**1.3 (1.06–1.58)**	**0.016**	0.508	**1.31 (1.07–1.58)**	**0.008**	0.528	1.31 (0.01–11.09)	0.861	0.508	1.89 (0.21–7.72)	0.497
Acute MI	0.523	**1.69 (1.43–1.96)**	**<0.001**	0.521	**1.64 (1.4–1.9)**	**<0.001**	0.541	0.87 (0.07–7.37)	0.923	0.541	0.39 (0–2.91)	0.445
Thyroid Disease	0.502	0.98 (0.87–1.1)	0.725	0.506	0.94 (0.83–1.05)	0.261	0.622	0.24 (0–2.02)	0.230	0.559	0.62 (0.12–2.07)	0.468
Vision Loss	0.509	1.11 (0.96–1.27)	0.203	0.506	1.1 (0.95–1.28)	0.242	0.605	0.71 (0.07–4.01)	0.807	0.568	0.32 (0–2.38)	0.297
Hearing Loss	0.504	0.96 (0.85–1.08)	0.501	0.503	0.97 (0.85–1.09)	0.587	0.612	0.27 (0–2.24)	0.273	0.541	0.69 (0.13–2.3)	0.577
Dementia	0.508	1.11 (0.99–1.26)	0.077	0.507	1.13 (0.98–1.3)	0.089	0.665	0.52 (0–4.36)	0.622	0.607	1.35 (0.26–4.5)	0.677
Melanoma	0.501	0.85 (0.43–1.45)	0.562	0.501	0.78 (0.42–1.34)	0.391	0.504	8.85 (0.07–76.84)	0.261	0.504	3.96 (0.03–24.82)	0.431
Blood Cancer	0.500	0.97 (0.87–1.34)	0.690	0.501	0.92 (0.84–1.29)	0.649	0.511	3.57 (0.03–20.27)	0.485	0.511	1.6 (0.01–11.97)	0.761
Parkinson's Disease	0.500	1.07 (0.7–1.56)	0.700	0.501	1.11 (0.78–1.6)	0.584	0.506	4.93 (0.04–41.69)	0.379	0.508	2.21 (0.02–16.51)	0.626
Liver Disease	**0.514**	**1.57 (1.21–2.02)**	**<0.001**	0.513	**1.54 (1.2–1.97)**	**<0.001**	0.555	0.97 (0–16.78)	0.826	0.516	0.11 (0–4.12)	0.311

Bold text indicates p < 0.05 (statistically significant).
AUC: 0.65–0.80=Good; >0.80=Excellent; <0.55=Poor discrimination
eFI Score modeled with 40 (5 knots), Other components modeled with unadjusted Firth's penalized logistic regression
AUC = area under ROC Curve; OR = Odds Ratio; CI = Confidence Interval;

Results: Mortality Discrimination (Limited Power)

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Overall eFI Performance

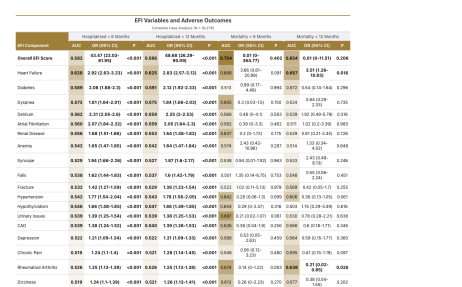
Mortality

- 6-month AUC = **0.754** (hosp 6mo AUC = 0.628)
- 12-month AUC = **0.654**

Caution: Point estimates may be unstable. Only 6 deaths at 6 months, 14 deaths at 12 months. Confidence intervals are extremely wide, statistical tests lack power

Top Individual Components

- Heart Failure (hosp 6mo AUC = 0.628)
- Diabetes, Dyspnea, Delirium



eFI Variables and Adverse Outcomes

Complete Case Analysis: N = 10,770

	Hospitalized < 6 Months			Hospitalized < 12 Months			Mortality < 6 Months			Mortality < 12 Months		
eFI Component	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P	AUC	OR (95% CI)	P
Overall EFI Score	0.580	**85.07 (23.03-313.93)**	**<0.001**	0.590	**48.68 (28.28-83.78)**	**<0.001**	**0.754**	**9.01 (2-44.27)**	**0.002**	**0.654**	**0.01 (<11.51)**	**0.206**
Heart Failure	0.628	2.92 (2.62-3.23)	<0.001	0.621	2.83 (2.57-3.12)	<0.001	0.650	3.80 (0.91-20.89)	0.081	0.657	**3.51 (1.26-10.83)**	**0.016**
Diabetes	0.589	2.08 (1.88-2.3)	<0.001	0.591	2.12 (1.92-2.33)	<0.001	0.613	0.99 (0.11-4.40)	0.994	0.672	0.54 (0.14-1.84)	0.298
Dyspnea	0.573	1.61 (1.44-2.01)	<0.001	0.575	1.64 (1.46-2.03)	<0.001	0.655	0.3 (0.03-1.51)	0.158	0.524	0.84 (0.29-3.33)	0.795
Delirium	0.562	2.31 (2.09-2.6)	<0.001	0.559	2.25 (2-2.53)	<0.001	0.598	0.48 (0-4.1)	0.583	0.539	1.02 (0.49-3.79)	0.976
Atrial Fibrillation	0.556	2.07 (1.84-2.32)	<0.001	0.559	2.05 (1.84-2.3)	<0.001	0.582	0.39 (0-3.3)	0.462	0.511	1.02 (0.2-3.39)	0.983
Renal Disease	0.556	1.68 (1.51-1.88)	<0.001	0.563	1.64 (1.48-1.82)	<0.001	0.637	0.2 (0-1.75)	0.175	0.639	0.81 (0.21-2.68)	0.739
Anemia	0.542	1.65 (1.47-1.85)	<0.001	0.542	1.64 (1.47-1.84)	<0.001	0.574	2.43 (0.42-10.96)	0.287	0.514	1.33 (0.34-4.02)	0.648
Syncope	0.529	1.96 (1.66-2.26)	<0.001	0.527	1.87 (1.6-2.17)	<0.001	0.639	0.84 (0.01-7.93)	0.963	0.633	2.43 (0.48-8.13)	0.249
Falls	0.538	1.82 (1.64-1.83)	<0.001	0.537	1.6 (1.42-1.79)	<0.001	0.581	1.35 (0.14-6.70)	0.753	0.548	0.99 (0.09-3.24)	0.451
Fracture	0.532	1.42 (1.27-1.58)	<0.001	0.529	1.39 (1.23-1.54)	<0.001	0.522	1.02 (0.11-5.13)	0.979	0.569	0.42 (0.05-1.7)	0.255
Hypertension	0.542	1.77 (1.54-2.04)	<0.001	0.543	1.78 (1.55-2.05)	<0.001	0.642	0.28 (0.06-1.3)	0.099	0.606	0.56 (0.13-1.95)	0.001
Hypothyroidism	0.546	1.66 (1.48-1.85)	<0.001	0.547	1.66 (1.48-1.85)	<0.001	0.604	0.29 (0-2.47)	0.316	0.503	1.15 (0.29-3.69)	0.816
Urinary Issues	0.539	1.39 (1.25-1.54)	<0.001	0.539	1.39 (1.25-1.53)	<0.001	0.597	0.21 (0.02-1.07)	0.061	0.539	0.70 (0.28-2.23)	0.538
CAD	0.538	1.38 (1.24-1.53)	<0.001	0.543	1.39 (1.26-1.53)	<0.001	0.639	0.38 (0.04-1.9)	0.256	0.566	0.8 (0.28-1.71)	0.548
Depression	0.522	1.21 (1.09-1.34)	<0.001	0.522	1.21 (1.09-1.33)	<0.001	0.555	0.53 (0.05-3.60)	0.459	0.564	0.59 (0.15-1.77)	0.360
Chronic Pain	0.519	1.24 (1.11-1.4)	<0.001	0.521	1.28 (1.14-1.45)	<0.001	0.548	0.56 (0.12-2.23)	0.480	0.599	0.41 (0.15-1.18)	0.097
Rheumatoid Arthritis	0.526	1.25 (1.13-1.39)	<0.001	0.526	1.25 (1.13-1.38)	<0.001	0.474	0.14 (0-1.22)	0.083	0.638	**0.21 (0.02-0.89)**	**0.028**
Dizziness	0.519	1.24 (1.1-1.39)	<0.001	0.521	1.26 (1.12-1.41)	<0.001	0.613	0.26 (0-2.23)	0.270	0.577	0.39 (0.04-1.56)	0.202
Valvular Disease	0.519	1.41 (1.22-1.62)	<0.001	0.517	1.38 (1.19-1.58)	<0.001	0.555	0.62 (0-5.23)	0.725	0.555	0.28 (0-2.07)	0.269
Cancer (Other)	0.515	0.86 (0.76-0.98)	0.009	0.516	0.86 (0.76-0.96)	0.004	0.547	0.77 (0.06-3.96)	0.773	0.618	0.56 (0.17-1.68)	0.281
Osteoporosis	0.518	0.72 (0.61-0.84)	<0.001	0.519	0.71 (0.6-0.82)	<0.001	0.568	0.49 (0-4.13)	0.589	0.568	0.22 (0-1.66)	0.174
Weight Loss	0.516	0.82 (0.72-0.93)	0.002	0.519	0.78 (0.68-0.88)	<0.001	0.551	0.3 (0-2.58)	0.332	0.539	0.79 (0.15-2.62)	0.721
Stroke	0.515	1.18 (1.01-1.36)	0.021	0.510	1.10 (1.03-1.36)	0.016	0.567	0.5 (0-4.19)	0.596	0.567	0.22 (0-1.66)	0.179
Peptic Ulcer	0.506	1.44 (1.1-1.95)	0.008	0.506	1.45 (1.12-1.88)	0.006	0.515	2.56 (0.62-21.67)	0.574	0.521	3.7 (1.4-15.75)	0.199
Peripheral Vascular Disease	0.513	1.13 (1.01-1.29)	0.028	0.513	1.12 (1.01-1.25)	0.027	0.572	0.81 (0.06-5.04)	0.872	0.548	0.68 (0.17-2.06)	0.511
Incontinence	0.507	1.3 (1.04-1.58)	0.016	0.508	1.31 (1.07-1.59)	0.009	0.538	1.31 (0.21-11.05)	0.861	0.508	1.89 (0.21-7.72)	0.497
Acute MI	0.523	1.08 (0.43-1.96)	<0.001	0.521	1.64 (1.4-1.9)	<0.001	0.541	0.67 (0.01-7.37)	0.923	0.541	0.39 (0-2.91)	0.445
Thyroid Disease	0.502	0.98 (0.87-1.1)	0.725	0.506	0.94 (0.83-1.05)	0.261	0.422	0.24 (0-2.02)	0.230	0.552	0.62 (0.12-2.07)	0.468
Vision Loss	0.505	1.11 (0.94-1.31)	0.203	0.504	1.1 (0.94-1.29)	0.242	0.508	0.71 (0.07-4.67)	0.807	0.508	0.33 (0-2.96)	0.337
Hearing Loss	0.504	0.96 (0.85-1.09)	0.581	0.503	0.97 (0.86-1.09)	0.587	0.512	0.27 (0-2.24)	0.273	0.541	0.69 (0.13-2.3)	0.577
Dementia	0.508	1.11 (0.95-1.28)	0.177	0.507	1.13 (0.96-1.3)	0.089	0.565	0.52 (0-4.38)	0.622	0.507	1.35 (0.26-4.5)	0.877
Melanoma	0.501	0.85 (0.67-1.07)	0.162	0.501	0.78 (0.42-1.34)	0.391	0.504	0.83 (0.07-15.34)	0.261	0.504	3.96 (0.03-29.42)	0.431
Blood Cancer	0.500	0.87 (0.67-1.36)	0.650	0.501	0.92 (0.84-1.23)	0.649	0.511	3.57 (0.83-20.27)	0.485	0.511	1.15 (0.01-11.97)	0.761
Parkinson's Disease	0.500	1.07 (0.7-1.56)	0.150	0.501	1.11 (0.79-1.6)	0.583	0.508	4.93 (0.84-41.69)	0.379	0.508	2.31 (0.02-16.31)	0.626
Liver Disease	**0.514**	**1.57 (1.21-2.02)**	**<0.001**	**0.512**	**1.64 (1.2-1.97)**	**<0.001**	0.555	0.67 (0-19.76)	0.825	0.555	0.11 (0-4.13)	0.311

Bold text indicates p < 0.05 statistically significant.

AUC (left) <50= chance; 60=fair; 70=good; 80=very good; 90=excellent

EFI Score modeled with AUC (5 levels). Other components modeled with unadjusted HFRS penalized logistic regression.

AUC = Area Under ROC Curve; OR = Odds Ratio; CI = Confidence Interval.

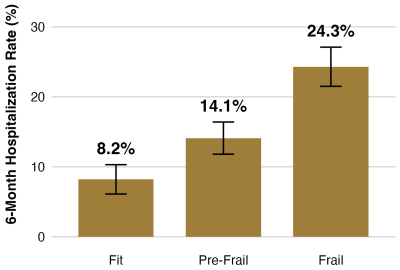
Results: Hospitalization Risk by Frailty

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Hospitalization Rates by Frailty	
Category	**6 Month Hospitalization**
Fit (≤ 0.10)	8.2%
Pre-Frail (0.10–0.21)	14.1%
Frail (> 0.21)	24.3%

Statistical Significance

- Clear gradient: $p < 0.001$
- Frail vs. Fit: OR = 2.13 (95% CI: 1.88–2.42)



6 month hospitalization rates increase across frailty category (error bars = 95% CI).

Results: RCS Predicted Probabilities

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RCS Predictions with Firth Penalization: Continuous eFI Score and Adverse Outcomes

		Hospitalization				Mortality			
eFI Percentile	eFI Score	6-mo Prob	OR vs 10th	12-mo Prob	OR vs 10th	6-mo Prob	OR vs 10th	12-mo Prob	OR vs 10th
10th	0.070	6.049608%	1.00	6.377143%	1.00	0.3025197%	1.00	0.3195852%	1.00
25th	0.112	7.842594%	1.32	8.358020%	1.34	0.2816990%	0.93	0.2723033%	0.85
50th	0.163	8.891486%	1.52	9.468676%	1.54	0.2387128%	0.79	0.2543891%	0.80
75th	0.218	10.506938%	1.82	11.437263%	1.90	0.2657802%	0.88	0.2093875%	0.65
90th	0.269	12.440430%	2.21	13.511561%	2.29	0.2776121%	0.92	0.2298960%	0.72
95th	0.302	13.813836%	2.49	14.854114%	2.56	0.2744705%	0.91	0.2690442%	0.84

Modeled with restricted cubic splines (5 knots) using Firth's penalized logistic regression.

Probabilities are predicted risks; ORs are relative to the 10th percentile eFI score.

Interpretation

Patients at the 95th percentile of frailty have approximately 3× the odds of 6-month hospitalization compared to those at the 10th percentile.

Limitations

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Missing Data

- 26.7% excluded due to missing eFI
- Excluded significantly older
- Selection bias possible

Event Sparsity

- Only 6 deaths at 6 months
- Only 14 deaths at 12 months
- Even Firth's underpowered

Methodological

- Unadjusted models
- No external validation
- Single-cohort analysis

Performance Gap Hospitalization AUC = 0.582 vs. Pajewski's ≈ 0.65. Missing utilization variables likely explains the difference.

Future Directions

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Machine Learning

- Random forest, XGBoost to improve AUC
- Target: match Pajewski's $+0.05$ AUC beyond age + CCI

Model Enhancement

- Add utilization variables (prior ED visits, admissions)
- Develop calibration plots
- Multivariable adjusted models

Clinical Implementation

Flexdashboard / Shiny prototype for bedside risk prediction

Thank You

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